Following the Card: Auditing Performance Attribution in Federated Tool Workflows

Chang Wang
Department of Electrical and Computer Engineering
University of Nevada, Las Vegas
Las Vegas, USA
changw11@unlv.nevada.edu

Ming Zhu
Department of Electrical and Computer Engineering
University of Nevada, Las Vegas
Las Vegas, USA
ming.zhu@unlv.edu

Shengjie Zhai
Department of Electrical and Computer Engineering
University of Nevada, Las Vegas
Las Vegas, USA
shengjie.zhai@unlv.edu

***Abstract*—Tool-agent systems increasingly expose a structured runtime record — a *workflow card* — reporting execution progress, verified identifiers, write eligibility, a controller's suggested next action, and answer readiness. The same record is used twice: to generate local training targets, and as part of what the deployed policy observes while acting. We show that this arrangement makes conventional closed-loop results tables unable to separate what a policy learned from what it was handed, and we audit a federated workflow-card system, FedWCD, across two studies. In a 20-client controlled environment, an exact rule over the same runtime view reaches 1.0000 task success while the best learned configuration reaches 0.9875: learning buys nothing over a rule consuming the same information. A preference-free supervised control sharing the same anchors, action labels, and modular structure reaches 0.9825, a gap of five parts in a thousand that cannot carry an attribution to the preference objective. Corrupting 5% of next-action labels leaves success at 0.9669 while the severity-weighted violation score rises tenfold, from 0.0125 to 0.1313, and a verifier-event decomposition over $1{,}600$ validation episodes attributes 9.3 of those 13.1 events per 100 episodes to sequence deviations rather than to unsafe writes. Withholding the next-action field drops success to 0.294 even though the remaining fields determine that action exactly. Across fifteen reported conditions, nine satisfy $V = 1 - S$ to within 0.001, so the compliance metric restates the success metric in most of them. Externally, four structurally distinct training configurations report the same accuracy to four decimals, and two-thirds of a $+0.372$ advantage we previously published over a generic control is a difference in how often each system commits to an answer: scoring ties at one half leaves $+0.122$. One advantage survives with an interval that excludes zero, and it is 2.95 points. We close with the environment and protocol changes that restore attribution.**

***Index Terms*—tool agents, performance attribution, evaluation methodology, workflow supervision, federated learning, benchmark design**

1. Introduction

A tool agent that changes a flight booking must locate the reservation, confirm the passenger is eligible, issue the write, and only then report what happened. Because a correct final answer can be reached through a wrong intermediate path, evaluation has moved toward scoring decisions throughout execution rather than only at the end [1], [2], [3], [4], [5]. Workflow stacks make this convenient: controllers, verifiers, and database validators already emit a structured per-step record, which we call a *workflow card*. A card carries execution progress, which identifiers have been grounded, whether a write is currently admissible, whether the workflow is ready to end, and the controller's suggested next action.

In the system we study, and in the designs it is built from, the card is used twice. It generates local training targets, and it forms part of what the deployed policy observes while acting. This paper asks a narrow question about that arrangement: given a conventional closed-loop results table, which performance differences can be attributed to which design choices?

Our own earlier reporting of this system answered that question too generously. We presented a modular federated configuration as outperforming its controls, and a structured-state ranker as substantially beating a generic baseline. Three observations from the same records overturn both readings. An exact rule over the same runtime view already solves the controlled environment. A preference-free supervised control lands within 0.005 of the full configuration. Four structurally distinct training configurations in the external ranking study report the same accuracy to four decimals. None of this means the system does not learn; each of it means the reported comparison does not isolate the component it was built to isolate.

We therefore reorganise the work as an attribution audit. Section 3 defines the condition at issue — an *action-revealing* runtime input, in which the controller's action label is available to the policy at execution — and separates it from the much weaker property that the input merely determines the target. Section 4 describes the two studies and the four comparisons the audit rests on. Sections 5 and 6 report them.

Section 7 specifies what has to change for the attribution questions to become answerable, and turns the negative result into a stated experiment.

Contributions. (i) A narrow definition of action-revealing runtime input, separated from informational sufficiency, with the conditions under which three commonly drawn consequences follow. (ii) An attribution audit of a federated workflow-card system built from four comparisons — a rule baseline over the same view, a card-noise perturbation, a runtime-view ablation paired with a step-level aliasing measurement, and a named-axis spread — reported for two studies. (iii) A correction to our own external ranking result: a published $+0.372$ advantage is two-thirds an artefact of scoring ties as errors, and we give the identity that lets any reader perform the same decomposition on a published table. (iv) A set of environment and protocol changes that restore attribution, with the one experiment that converts this paper's negative finding into a well-posed learning problem.

2. Related Work

Attribution and measurement validity. Reported gains are attributable only to the extent that comparisons isolate the intended axis. Dataset bias [6], hypothesis-only baselines [7], shortcut learning [8], and benchmark validity in language understanding [9] all document how easily a benchmark rewards something other than the capability it names. Here the feature at issue is a runtime field supplied deliberately by the system design rather than a statistical correlate found in a static corpus. That makes it directly manipulable: the experimenter can corrupt it or withhold it, which is what the comparisons of Section 4 do.

Privileged information and imitation. Learning using privileged information formalises the gap between what a teacher may use and what a student may use at test time [10]; imitation learning studies the distribution shift a learner induces on itself [11]. Card-conditioned agents sit awkwardly in that framing, because a field intended as the teacher's is also handed to the policy at execution. Whether any of the teacher's mapping reached the parameters is then an empirical question, and Section 5.3 measures it.

Tool-agent evaluation. AgentBench, WebArena, OS-World, and safety-oriented web-agent benchmarks score intermediate actions through verifiers and domain rules [2], [3], [4], [5]. The τ-bench family supplies tool–agent–user settings, with Airline and Retail revisions documented by the project [12], [13]. These benchmarks differ in how much structured state they expose to the agent, and the amount exposed is what decides whether a rule baseline over that state is informative.

Federated and preference-based training. FedAvg and its successors study decentralised optimisation under client heterogeneity [14], with agnostic federated learning motivating tail-client reporting [15]; parameter-efficient and federated adaptation address update representation [16], [17]; instruction tuning, direct preference optimisation (DPO), and federated preference methods supply feedback-based objectives [18], [19]. The system audited here draws on all of these, and the audit asks whether our experiments separate their contributions.

3. Methodology

3.1. Setting and notation

Let N clients be indexed by $i \in \{1, \dots, N\}$, each owning its tasks, tools, workflow state, and verifier. At execution step t, let e_t be the private environment state, g_t the full workflow card, and b_t the base execution feedback: task type, execution progress, the previous action, counters, and available verifier feedback. A view map R selects which card fields the deployed policy observes:

$$z_t = R(g_t), \qquad x_t = (b_t, z_t), \tag{1}$$

so z_t is the retained card view and x_t the policy's input. Let θ denote the policy parameters and $\pi_\theta(a \mid x)$ the probability of action a given input x; an action is a tool choice or an answer/stop decision. A local teacher $\mathcal{T}_i$ maps private state to a target, $y_t = \mathcal{T}_i(e_t, g_t)$. An *episode* is one task execution; a trajectory of length T is $\zeta = ((x_t, a_t, o_t))_{t=1}^{T}$, with o_t the tool or environment response. Table 1 lists the card fields and both of their uses.

3.2. Action-revealing inputs, and what they are not

Definition 1 (Action-revealing view)**.** *Fix a state distribution and a target map $\mathcal{T}_i$. A view R is* action-revealing *at step t if the target action y_t appears verbatim as a component of z_t, or is recoverable from z_t by a decoding rule of constant description length fixed in advance of training.*

This is narrower than the property that y_t is *some* function of z_t. That weaker property — informational sufficiency — holds in most well-posed supervised problems and carries no evidential weight on its own: a learner that must compute a hard function of its input is doing the work we want it to do. Definition 1 isolates the case where no non-trivial computation stands between input and target. In the environment studied here the next-action field satisfies it by construction, because it is the controller's action label placed directly into the card.

Three consequences are commonly drawn from this situation. Each holds under further conditions, which we state rather than assume.

(C1) A rule attains the ceiling. If R is action-revealing and the teacher's action is optimal for the evaluated objective, then the deterministic policy $\pi_{\mathrm{rule}}(z) = y$ is optimal. The additional conditions are teacher optimality, executable tool arguments, and reachability of the intended states under closed-loop execution. In our environment a rule reaches 1.0000 (Section 5.1), which we report as a measurement.

(C2) Field corruption propagates into the violation score. A policy that executes the field inherits its errors. The magnitude depends on how violations are weighted, how many steps are scored, and whether a wrong action induces further violations downstream. We therefore report

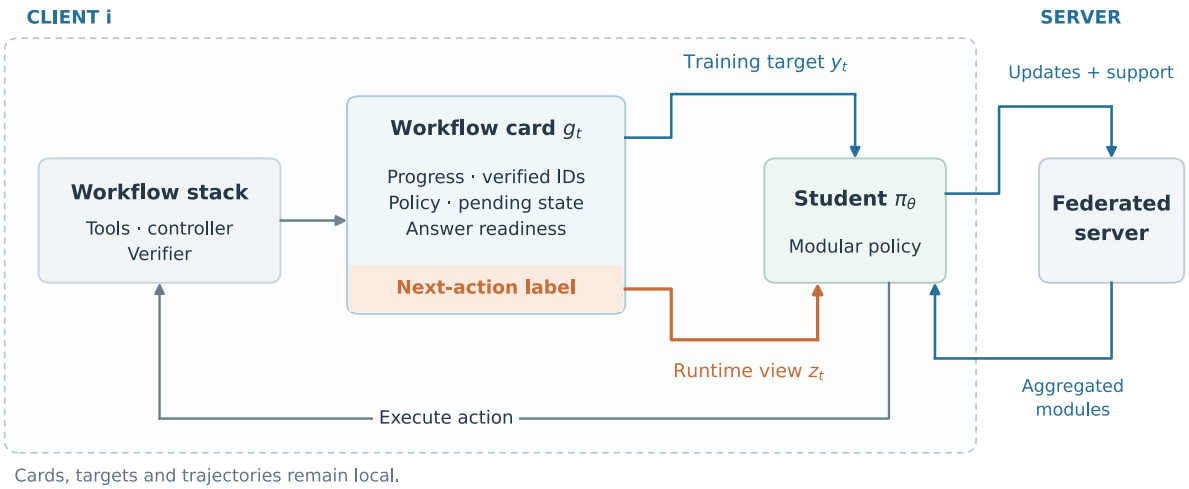


Figure 1. The two uses of a workflow card. It generates the local training target y_t, and it supplies part of the runtime view $z_t = R(g_t)$ that the policy observes while acting. When the controller's next-action field is retained in z_t (highlighted), the training target is available to the policy at execution time. Only module parameter changes and support statistics cross the client boundary.

TABLE 1. WORKFLOW-CARD FIELDS AND THEIR TWO USES. "TARGET" MEANS THE FIELD PARTICIPATES IN GENERATING A SUPERVISION SIGNAL; "RUNTIME INPUT" MEANS IT IS VISIBLE TO THE POLICY WHILE ACTING. THE LAST COLUMN MARKS ONLY THE CASE OF DEFINITION 1: THE TARGET *action* IS ITSELF PRESENT IN THE RUNTIME VIEW. A FIELD THAT MERELY HELPS DERIVE A TARGET IS NOT MARKED.

Field	Local meaning	Use as target	Use as runtime input	Action-revealing?
Progress / evidence	Required reads or checks completed	Action targets	Progress input	no
Verified identifiers	Identifiers matched to the record	Grounding, verifier checks	Grounding input	no
Policy / eligibility	Conditions for a permitted write	Admissibility labels	Admissibility input	no
Pending write	Backend change not yet committed	Completion guidance	State input	no
Next action	**Controller's suggested decision**	**Action cross-entropy**	**Full-card input**	**yes**
Answer readiness	Whether the workflow may end	Stop/answer labels	Full-card input	for the stop decision

the measured sensitivity (Section 5.2) as a sensitivity, not as a copy rate.

(C3) Compliance becomes redundant with success. If the evaluator scores deviation from a single canonical sequence, and a failing episode contributes about one unit of severity-weighted violation, then the violation score and the failure rate track each other. Both premises are environment-specific; the redundancy appears in nine of fifteen reported conditions (Section 5.5).

One terminological point. Using a legitimately available runtime field is neither a train/test leak nor a privacy leak. We call the phenomenon *runtime-information dependence* and reserve "leakage" for its established meanings [20], [21], [22].

4. Experimental Setup

4.1. System under audit

FedWCD (Federated Workflow-Card Distillation, the name used in our earlier records) trains each client from three local sources: verifier-approved example trajectories, called *anchors*, fitted by supervised fine-tuning (SFT); card-derived next-action labels fitted by action cross-entropy (ActionCE); and same-task trajectory preferences fitted with a DPO-style objective at preference scale β_{pref}. The policy is modular over workflow functions: read/extract, eligibility/check, write/commit, answer/stop, and recovery. For module k at communication round r, let $\mathcal{I}_{k,r}$ be the clients that activated it; each uploads a parameter change $\Delta\theta^r_{i,k}$ and a support statistic $s^r_{i,k}$, and the server aggregates with support clipped at $s_{\max}$ and server step size η_s,

$$\bar{s}^r_{i,k} = \text{clip}(s^r_{i,k}, 0, s_{\max}), \qquad w^r_{i,k} = \frac{\bar{s}^r_{i,k}}{\sum_{j\in\mathcal{I}_{k,r}} \bar{s}^r_{j,k}},$$
$$\theta^{r+1}_k = \theta^r_k + \eta_s \sum_{i\in\mathcal{I}_{k,r}} w^r_{i,k}\, \Delta\theta^r_{i,k}, \tag{2}$$

leaving the module unchanged when the denominator is zero. The policy backend in all ToolWorld-v2 experiments is a compact vector policy over a fixed feature encoding, not a language model. Raw trajectories, cards, tool returns, and anchors remain local.

TABLE 2. TOOLWORLD-V2 CONFIGURATION. A COMMUNICATION ROUND IS ONE FEDERATED UPDATE CYCLE.

Setting	Value
Clients / participation fraction	20 / 0.5
Communication rounds	20
Training seeds	42–46
Local tasks / client / round	32
Rollouts per local task	4
Local preference pairs	32 / client / round
Validation / held-out tasks	16 / 32 per client
Full anchor coverage	32 trajectories / client
Local learning rate / epochs	0.003 / 1
Preference scale β_{pref}	0.1
Mixed / continued SFT weights	0.15 / 0.25
ActionCE weight	0.35
Scoring weights: violation, call, tool, irrelevant, irreversible	2.0, 0.2, 0.025, 0.25, 2.0

4.2. ToolWorld-v2

ToolWorld-v2 is a controlled workflow environment with eight task types, 12 tools, four answer classes, and 20 clients differing in task-type distribution and long-tail tool coverage, giving a non-IID split. The policy acts for at most ten steps per episode; reference plans contain two to four steps. Tool noise is 0.10 and the irreversible-operation probability is 0.15. Table 2 lists the training configuration.

Metrics are task success S, the fraction of held-out episodes solved; the severity-weighted violation score V per episode; the mean number of tool calls C per episode; and worst-client success W, the minimum of per-client success. Safe success is the fraction of episodes that succeed while meeting the evaluator's per-episode constraints; required-tool recall (RTR) is coverage of the tools a workflow requires; UnderCall and NoOp are the fractions of episodes failing through insufficient tool use and through no-operation behaviour. A checkpoint is *feasible* when $V \leq 0.05$, $C \leq 3.5$, $S \geq 0.20$, safe success ≥ 0.05, $W \geq 0.05$, $\text{RTR} \geq 0.50$, $\text{UnderCall} \leq 0.50$, and $\text{NoOp} \leq 0.50$. Validation selects checkpoints; tables report held-out results.

The verifier records inadmissible writes, incorrect or stale final responses, and deviations from the prescribed next-action sequence. The third category dominates (Section 5.6), so V is principally a workflow-conformance measure with a small safety component, and is not a count of independently established business-safety incidents.

4.3. External τ^3 Airline offline ranking

The external study is an offline ranking diagnostic built from τ^3 Airline contexts: a linear ranker chooses between two fixed candidate responses or tool actions drawn from the same source context. The original collection has 2,060 pairs and 4,120 candidates, split into 1,260 training, 400 selection, and 400 audit pairs, with no source-state overlap across splits. A later challenge set probes verified identifiers, policy/write state, and pending/answer readiness with 600/120/120 pairs per group, giving 2,520 pairs of which 360 are audit pairs.

Labels are simulated rather than human. A visible-state oracle determines the preferred candidate from the available fields; in the E75 tier about 75% of labels agree with that oracle, and in the E50 tier about 50%, providing a random-feedback check. Five label seeds vary the synthetic labels; they are not five independent collections of Airline tasks. The ranker's features are drawn from the same visible state that generates the labels, so the E75 agreement rate is the expected accuracy of an oracle-equivalent scorer under the stated noise model.

4.4. The four comparisons

The audit rests on four comparisons. We label them A1–A4 and use those labels as signposts in Sections 5 and 6. Each isolates a different way in which a results table can fail to attribute performance, and each has a scope that we state where it is used.

A1, rule comparison. Let π_{rule} be the exact deterministic policy that reads the runtime view and executes the action it indicates, using no information the learned policy lacks. The statistic is the rule gap $\Delta_{\text{rule}} = S(\pi_\theta) - S(\pi_{\text{rule}})$. A non-negative rule gap is the minimum evidence that learning contributed something on the evaluated distribution.

A2, card-noise sensitivity. Corrupt the next-action field with independent noise at rate ε during training and execution, while the evaluator continues to score against the clean canonical sequence, and measure $V(\varepsilon)$. This separates a policy whose scored decisions depend on the controller's labels from one that would survive their degradation.

A3, runtime-view ablation. Remove fields from the view and measure episode success, alongside a step-level measurement of the same views: the *aliasing rate*, the fraction of full states that share a masked cell with a differing action label, and *Step-BayesErr*, the induced lower bound on single-step action-label error. The two scales are reported side by side and never combined — single-step label accuracy bounds a one-step classification problem, whereas episode success spans several actions and permits recovery.

A4, named-axis spread. Take the spread of reported scores across configurations that differ along one named axis. A spread smaller than the differences the same table resolves elsewhere means the experiment does not separate that axis. Naming the axis is part of the measurement: a spread taken over the two most similar configurations would say nothing about the environment as a whole.

5. Audit I: ToolWorld-v2

5.1. A1 — learning buys nothing over a rule on the same view

Table 3 reports the main results. The exact card rule — which reads the same card the policy sees and executes the indicated action — solves every evaluated task with zero violations and perfect worst-client success. The best learned configuration reaches 0.9875, so $\Delta_{\text{rule}} = -0.0125$.

TABLE 3. MAIN TOOLWORLD-V2 RESULTS. MEANS OVER SEEDS 42–46 AFTER VALIDATION-BASED CHECKPOINT SELECTION. THE RULE ROW IS NOT A BASELINE IN THE USUAL SENSE: IT CONSUMES THE SAME RUNTIME VIEW AS THE LEARNED ROWS AND BOUNDS THEM FROM ABOVE. DASHES MARK QUANTITIES ABSENT FROM THE CURRENT EXPERIMENT GENERATION.

Method or condition	$S\uparrow$	$V\downarrow$	$C\downarrow$	$W\uparrow$	Feas.
Exact card rule, same view	1.0000	0.0000	—	1.0000	5/5
FedWCD, full modular config.	0.9875	0.0125	2.44	0.8875	5/5
Anchor SFT + ActionCE	0.9825	0.0175	—	0.8875	5/5
FedBiscuit-style control	0.9600	0.0400	—	0.7750	4/5
Anchor-only SFT	0.9463	0.0537	—	0.8000	4/5
Full objective, no anchors	0.8187	0.1812	—	0.4125	0/5
Single-block policy	0.7075	2.6938	—	0.3375	0/5

On this task distribution and under this evaluator, the learned policy is a lossy approximation of a rule that consumes the same runtime information. Every comparison in Table 3 is therefore a comparison among approximations to a known function, and the strong closed-loop numbers in it carry no information about whether workflow behaviour was learned.

5.2. A2 — compliance tracks the controller's label quality

Table 4 perturbs the card and Fig. 2 plots the label-noise conditions. Corrupting 5% of next-action labels leaves success nearly intact at 0.9669 while the violation score rises from 0.0125 to 0.1313, a factor of 10.5 that carries the policy well past the $V \leq 0.05$ feasibility budget. Removing 20% of verified identifiers or evidence reduces success to 0.8375 and raises violations to 0.1625.

The 5% condition separates success from compliance: most tasks still complete, because a single wrong call is recoverable inside a ten-step budget, while each corrupted label produces a conformance event. Compliance therefore tracks the quality of the controller's labels far more tightly than success does. A results table reporting success alone would show a 2-point change where the compliance metric shows a tenfold one.

We report the measured points and no fitted slope. A slope estimated from these data would not identify a rate at which the field is executed: the number of scored steps is not the number of tool calls, violations are severity-weighted rather than one unit per wrong action, and the policy is retrained at each noise level. An earlier experiment generation records $S = 0.959$ and $V = 0.242$ at $\varepsilon = 0.10$; because that generation reports $V = 0.029$ at $\varepsilon = 0$ against the current 0.0125, Fig. 2 keeps the two series separate.

5.3. A3 — the policy does not reconstruct a field it is not handed

Table 5 varies the runtime view. This block has its own full-card reference of 0.979 and is compared only

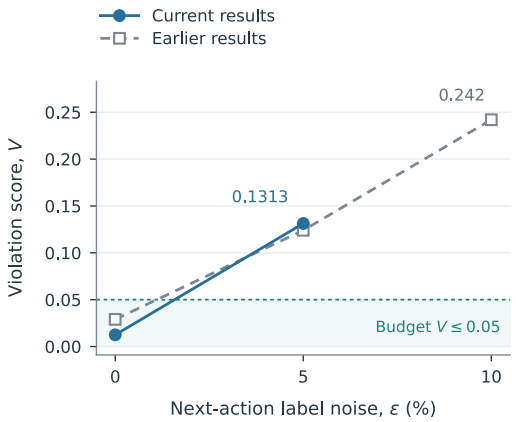


Figure 2. Card-noise sensitivity, with the two experiment generations kept separate. Solid: the current generation, $V = 0.0125$ at $\varepsilon = 0$ and 0.1313 at $\varepsilon = 0.05$. Dashed: an earlier generation, which also records $\varepsilon = 0.10$. The two series disagree at $\varepsilon = 0$ and are not pooled; no functional form is fitted to either, and the lines join points only. The shaded band is the feasibility budget $V \leq 0.05$, which 5% label noise already exceeds while task success remains at 0.9669.

TABLE 4. CARD-QUALITY PERTURBATIONS, CURRENT EXPERIMENT GENERATION. THE CLEAN ROW IS THE FEDWCD ROW OF TABLE 3.

Card condition	$S\uparrow$	$V\downarrow$	V/V_{clean}
Clean card	0.9875	0.0125	1.0
5% next-action label noise	0.9669	0.1313	10.5
20% missing verified IDs / evidence	0.8375	0.1625	13.0

within itself. Removing answer readiness preserves required-tool recall at 0.980 while success falls to 0.837 and worst-client success to 0.475, localising the loss to stopping and finalisation rather than to tool use. Removing the next-action field reduces success to 0.294; hiding all explicit card fields gives 0.261; a veto-only local gate does not recover the latter, at 0.252, as expected of a mechanism that can block an action but cannot supply missing state.

The step-level aliasing measurement turns the last two rows into the sharpest result in this block. Under base observation the masked view does not determine the action label at all: the aliasing rate is 1.000 and Step-BayesErr is 0.717, so the field was carrying the answer and nothing else in that view recovers it. Under the no-next-action view the remaining fields determine the action label exactly — aliasing 0.000, Step-BayesErr 0.000 — and episode success is still 0.294. Whatever these policies took from the card, it was not a mapping that reconstructs the next action from the fields that fix it.

That is the finding the word *distillation*, which we used for this system in earlier work, was meant to describe, and it is the finding these experiments do not support.

TABLE 5. RUNTIME-VIEW ABLATION WITH A STEP-LEVEL ALIASING MEASUREMENT. S, V, W, AND RTR ARE EPISODE-LEVEL; ALIAS AND STEP-BAYESERR ARE STEP-LEVEL STATISTICS OF THE ACTION LABEL UNDER THE SAME VIEW, COMPUTED CONDITIONING ON z_t WHILE THE POLICY ADDITIONALLY OBSERVES b_t. THE TWO GROUPS ARE ON DIFFERENT SCALES AND NO RATIO BETWEEN THEM IS FORMED. POLICIES ARE TRAINED FROM A FULL-CARD TEACHER AND EVALUATED UNDER THE LISTED VIEW. THIS BLOCK HAS ITS OWN FULL-CARD REFERENCE AND IS NOT COMPARED TO TABLE 3.

Policy view	Local gate	$S\uparrow$	$V\downarrow$	$W\uparrow$	RTR↑	Alias	Step-BayesErr
Full card	none	0.979	0.021	0.887	0.995	0.000	0.000
No answer-readiness	none	0.837	0.163	0.475	0.980	0.199	0.103
No next-action field	none	0.294	6.566	0.025	0.635	0.000	0.000
Base observation	none	0.261	6.570	0.025	0.639	1.000	0.717
Strict no-card	none	0.025	3.625	0.000	0.512	1.000	0.717
Base observation	veto only	0.252	6.378	0.013	0.647	1.000	0.717

5.4. A4 — the preference axis is unresolved; other axes are not

On the preference-objective axis, the full configuration and a control sharing its anchors, action labels, and modular structure differ by 0.0050 in success, with identical worst-client success at 0.8875 and identical feasible-seed counts. Five parts in a thousand, over five seeds, cannot carry an attribution to the preference channel — which is the one component distinguishing this system from straightforward imitation plus action supervision.

Other axes in the same table are resolved, and by margins two orders of magnitude larger. Across the six learned configurations of Table 3 the spread is 0.2800. Removing anchors costs 17 points of success and 48 points of worst-client success and leaves no feasible seed; replacing the modular parameterisation with a single block costs 28 points and raises the violation score from 0.0125 to 2.6938. The environment discriminates among design choices; it does not discriminate among these two.

The modular versus single-block contrast changes the policy parameterisation, so it attributes the difference to that parameterisation and not to federated aggregation. The matched centralised comparison that would separate the two is listed in Section 7.

5.5. Compliance restates success in most conditions

Of the fifteen distinct conditions in Tables 3–5, nine satisfy $|V - (1 - S)| \leq 0.001$; they are enumerated in the caption of Fig. 3. The six exceptions are the 5% noise condition and the five in which the policy has collapsed.

In the near-feasible region of these experiments the violation threshold therefore almost never binds independently of the success threshold, and reporting both conveys little more than reporting one. This is consequence (C3) of Section 3 realised in a specific environment, not a property of violation metrics in general: the 5% noise condition is a counterexample within our own data, where success and compliance carry different information, and the feasibility rule retains two independent constraints in the tool-call budget $C \leq 3.5$ and the utility floors.

TABLE 6. VERIFIER-EVENT DECOMPOSITION PER 100 VALIDATION EPISODES (1,600 EPISODES ACROSS SEEDS 42–46, FINAL ROUND). UNSAFE WRITES HAVE SEVERITY WEIGHT TWO.

Condition	Wrong-step	Unsafe wr.	Bad final	Under/over-call*	Total V
FedWCD	0.0	0.0	1.2	0.7 / 0.0	1.2
SFT+ActionCE	0.0	0.0	1.8	0.9 / 0.0	1.8
Single-block	260.9	5.9	2.6	28.6 / 32.8	269.4
5% card noise	9.3	0.5	3.3	2.6 / 5.8	13.1
Base observation	639.6	6.5	10.9	60.8 / 71.0	657.0

*non-additive diagnostics, reported for context only.

5.6. What the violations consist of

Table 6 decomposes the verifier diagnostics over the $1{,}600$ validation episodes underlying the corresponding rows. Next-action deviations account for 9.3 of the 13.1 weighted events per 100 episodes under 5% label noise, 260.9 of 269.4 for the single-block policy, and 639.6 of 657.0 under base observation. Unsafe irreversible writes, weighted at severity two, contribute 0.5, 5.9, and 6.5 respectively.

The measure is dominated by conformance to the controller's prescribed sequence in every non-trivial condition, so V measures agreement with one canonical path. An evaluator admitting alternative valid paths would produce a different and more informative quantity; Section 7 lists that change.

6. Audit II: τ^3 Offline Ranking

6.1. A4 — four training structures report one number

Table 7 reports training-structure controls on the challenge set. A full global/local configuration, centralised training, a global-head-only variant, and uniform modular aggregation all report 0.7444 to four decimals. The axes these configurations vary — whether training is federated at all, how clients are weighted, whether per-type heads exist — are not separated by this measurement.

The construction of the study explains the outcome. The labels were generated by a visible-state oracle from the available fields, and the ranker's features come from that same visible state. Measured full-view accuracies are 0.7444 overall and 0.7500, 0.7500, 0.7333 on the three field

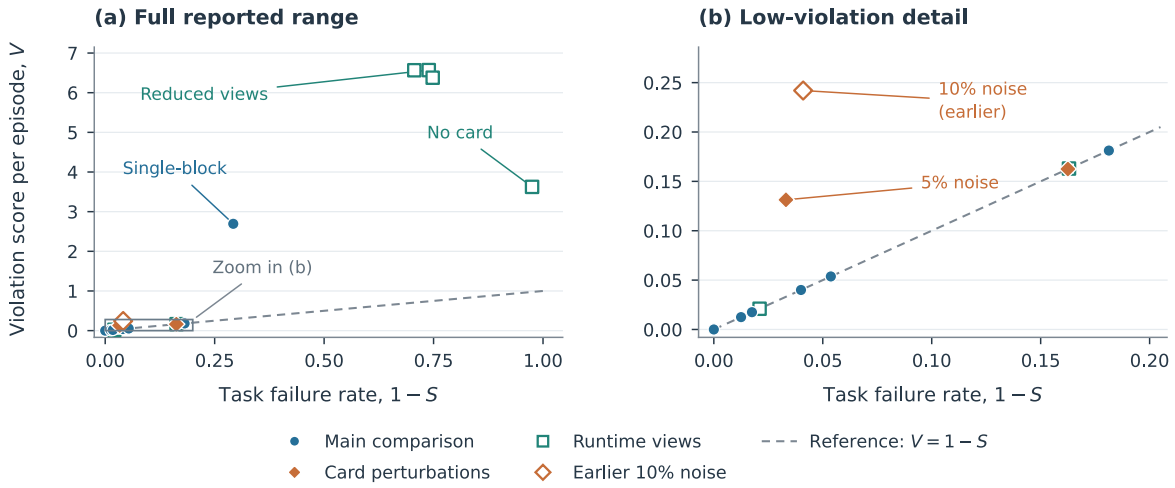


Figure 3. The violation score against the failure rate, by experiment block. Panel (a) shows the full reported range; panel (b) magnifies the low-violation region. Nine of the fifteen conditions in the current experiment generation satisfy $|V - (1 - S)| \leq 0.001$ and lie on the dashed line: the card rule, the full configuration, SFT+ActionCE, the FedBiscuit-style control, anchor-only SFT, the no-anchor configuration, the 20% missing-ID condition, and the full-card and no-answer-readiness runtime views. Six do not: the 5% label-noise condition and the five collapsed ones. The open marker is the $\varepsilon = 0.10$ point from the earlier experiment generation, plotted for context and excluded from the count for the reason given in Section 5.2. Markers distinguish the three experiment blocks, which are not pooled.

TABLE 7. TRAINING-STRUCTURE CONTROLS ON THE CHALLENGE SET: 360 AUDIT PAIRS, E75 LABELS, MEAN OVER FIVE LABEL SEEDS.

Training configuration	Reported accuracy
Full global/local configuration	0.7444
Centralised training	0.7444
Global head only	0.7444
Uniform modular aggregation	0.7444

TABLE 8. CHALLENGE-PROTOCOL ACCURACY UNDER ALTERNATIVE TIE-SCORING RULES (E75 LABELS). COMMITMENT RATES ARE RECOVERED USING EQ. (3).

System, E75	Ties as errors	Commitment	Ties at $\frac{1}{2}$
Structured state, full view	0.7444	≈ 1.000	0.7444
Best generic control	0.3722	0.5000	0.6222
No-card control	0.1222	0.1666	0.5389
Difference, full − generic	+0.3722	—	+0.1222

banks, against an E75 oracle agreement rate of 0.75, while E50 returns 0.5000. Every configuration approximates the oracle to within the label noise, which leaves the training-structure axis nothing to separate. Equal totals are not equal predictions, and the per-pair agreement of these four configurations is a separate quantity from their accuracy.

6.2. Two-thirds of a published gap is a scoring rule

Our earlier report of this study stated that the structured-state ranker reached 0.7444 against 0.3722 for the best generic control and 0.1222 for a no-card control, a +0.3722 advantage. The challenge protocol scores ties as errors. Under

TABLE 9. ORIGINAL AIRLINE PROTOCOL: 400 HELD-OUT AUDIT PAIRS, E75 LABELS, MEAN OVER FIVE LABEL SEEDS.

Ranker or comparison	Reported accuracy
Structured-state linear ranker	0.7125
Best predefined generic control	0.6830
Paired difference	+0.0295
95% CI, paired bootstrap grouped by source state	[+0.0135, +0.0450]

that rule, and provided the reversal control is an exact inversion of the same fixed scores on the same pairs with the same invalid handling, a system's reported accuracy and its reversed accuracy sum to its *commitment rate*, the fraction of pairs on which its score difference is non-zero:

$$\text{acc} + \text{acc}^{\text{flip}} = \Pr[\,\text{score difference} \neq 0\,]. \qquad (3)$$

Applying (3) to our own reversal figures recovers a commitment rate of 0.5000 for the generic control and 0.1666 for the no-card control (Table 8). The controls were not being outperformed so much as abstaining.

Scoring ties at one half instead of as errors moves the generic control from 0.3722 to 0.6222 and the advantage from +0.3722 to +0.1222. Two-thirds of the gap we published measured how often each system committed to an answer, not how often it was right. What the structured fields buy on this protocol is coverage.

Two corrections follow. Under a ties-as-errors rule the reversal control carries no independent information, because (3) fixes it once accuracy and commitment rate are known; we presented it as evidence against shortcut exploitation, and it is not. And a field-ablation accuracy of 0.0000 is produced both by a scorer that has become constant and

TABLE 10. CHANGES THAT RESTORE ATTRIBUTION, AND THE COMPARISON EACH SERVES. EXPECTED EFFECTS ARE PREDICTIONS; VALIDATING ONE OF THEM IS THE NEXT EXPERIMENT.

Change	What it alters	Serves	Expected effect
Withhold the next-action field at execution, retain it as a training target	The field supervises training but is absent from z_t	A1, A3	The card rule is no longer executable at test time, so A1 is re-posed against a rule using only the remaining fields; A3 becomes the primary measurement and tests amortisation directly
Intermittent or lagged cards	The field is present at a random subset of steps, or delayed by one step	A2, A3	Forces partial internalisation; card-noise sensitivity falls as field availability falls
Admit alternative valid paths	The evaluator accepts any compliant sequence	Sec. 5.5	Breaks the observed $V \approx 1-S$ redundancy and makes compliance an independent measurement rather than a restatement of failure
Matched centralised comparison	The same modular parameterisation trained centrally	A4	Separates the parameterisation effect of Section 5.4 from federated aggregation
Held-out workflow types	Evaluation on task types absent from local training	A1, A4	Separates configurations that generalise from configurations that fit; a hand-written rule is most likely to lose its advantage here
Report a rule baseline	$S(\pi_{\mathrm{rule}})$ over the same view, alongside every learned result	A1	Makes Δ_{rule} visible at the cost of one implementation
Report commitment rate and tie rule	Strict accuracy, commitment rate, and tie-at-$\frac{1}{2}$ accuracy together	Sec. 6.2	Removes the ambiguity between coverage and accuracy in forced-choice ranking
Labels independent of the audited scorer	Preferences from held-out outcomes or human annotation rather than from an oracle over the scorer's own features	A1, A4	Removes the shared-oracle ceiling that leaves the training-structure axis nothing to separate

by one that is systematically inverted, so that value does not identify which. Breaking ties is itself a property of the structured features: under a protocol that defines strict ranking in advance the original numbers are arithmetically correct, and what changes is their interpretation and their comparability across protocols.

Equation (3) is reusable under its stated conditions: a forced-choice evaluation that publishes a reversal control publishes its commitment rate implicitly, so any reader can perform this decomposition on a published table without access to code or data. Such evaluations should report commitment rate, strict accuracy, and tie-at-one-half accuracy together.

6.3. What survives: 2.95 points

On the original 400-pair audit the structured-state linear ranker reaches 0.7125 against 0.6830 for the best predefined generic control, a difference of 2.95 points with a source-state-grouped paired-bootstrap interval of $[+1.35, +4.50]$ points (Table 9). This is the larger and independently constructed of the two audit splits, and the generic control it is measured against scores 0.6830 — close to the 0.6222 that the challenge-protocol control reaches once its ties are scored at one half.

This is the only comparison in either study whose interval excludes zero, and it is the one advantage the audit leaves standing: structured state improves offline candidate ranking by about three points. It is a smaller and narrower claim than the one we published, and it is the claim the data support.

7. What Restores Attribution

Table 10 lists changes and the comparison each serves. The organising idea is to separate the target from the runtime view without discarding the structured state that makes these environments realistic.

The first row is the one that converts this paper's negative result into a research question. Training on the card while withholding it at execution is the setting in which absorbing the controller's mapping into parameters is both necessary and measurable, and Section 5.3 shows that the present policy does not absorb it even when the remaining fields determine the answer. Whether a policy with a richer history representation, a recurrent state, or an auxiliary field-reconstruction objective succeeds under that condition is the question these experiments pose.

8. Limitations

Statistical strength. Five training seeds resolve the differences of 17 points and more in Table 3. They do not resolve the 0.0050 preference-axis difference, and we draw no conclusion from it beyond its size. Paired intervals are reported for the external ranking comparison of Section 6.3 and for no other comparison.

Two experiment generations. Two generations of experiments contributed to these records and disagree on the no-anchor ablation, the eight-anchor condition, and the clean-card baseline. Every table here is drawn from a single generation. The one earlier record displayed anywhere in the paper, the $\varepsilon = 0.10$ point, is marked as such in Fig. 2 and excluded from every count.

Mask scope. Table 5 measures policies trained from a full-card teacher and evaluated under each listed view. It therefore reports what these policies do when a field is withheld, and does not separate that from what a policy trained under the mask could achieve.

Scope of the audit. The four comparisons address whether specific performance differences are attributable.

They do not address reward hacking, evaluator correctness, contamination, or whether the task suite measures anything of operational interest. They have been applied to one system.

Backend. All ToolWorld-v2 results use a compact vector policy over a fixed feature encoding; they are not evidence about language-model agents. The external study trains a linear ranker rather than any part of the ToolWorld policy.

Privacy. Keeping records local does not protect the information carried in module updates and support statistics; in a sparse-module design, which modules a client activates reflects which tool families it uses. Secure aggregation and differential privacy require their own implementation and utility evaluation [20], [21], [22].

9. Conclusion

A workflow card is used twice: to generate training targets, and as part of what a deployed policy observes. We asked what a conventional closed-loop results table can attribute under that arrangement, and audited our own federated workflow-card system across two studies. An exact rule over the same runtime view reaches 1.0000 success where the best learned configuration reaches 0.9875. A preference-free supervised control lands within 0.0050, too close to attribute anything to the preference objective. Corrupting 5% of next-action labels raises the violation score tenfold while success moves two points, and withholding the next-action field drops success to 0.294 even where the remaining fields determine that action exactly — the policy never learned to reconstruct what it was handed. Nine of fifteen conditions satisfy $V = 1 - S$ to within 0.001, so the compliance metric restates the success metric. Externally, four training structures report one number, two-thirds of a published $+0.372$ advantage is a tie-scoring rule, and a 2.95-point advantage for structured state survives with an interval excluding zero. None of this shows that the system fails to learn; it shows that the comparisons we built do not isolate the components they were meant to isolate. Section 7 states what restores them, beginning with withholding the controller's field at execution while keeping it as a training target — which turns the open question here into a well-posed one.

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